MIL-H-235C
23 February 1984
SUPERSEDING
MIL-H-235B
6 September 1977

MILITARY SPECIFICATION

HEXACHLOROETHANE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers technical grade hexachloroethane.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
UU-S-48	-	Sacks, Shipping, Paper
PPP-D-705	-	Drum, Shipping and Storage: Steel, 16 and 30 Gallon Capacity

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing - Interior Packaging

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:47Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1168 - Ammunition Lot Numbering

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applciable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

ASTM STANDARDS

D1193 - Reagent Water
D2989 - Acidity - Alkalinity of Halogenated Organic Solvents and Their Admixtures
E11 - Wire-Cloth Sieves For Testing Purposes
E203 - Water Using Karl Fischer Reagent
E256 - Chlorine in Organic Compounds by Sodium Peroxide Bomb Ignition
E324 - Relative Initial and Final Melting Points and the Melting Range of Organic Chemicals

Source: https://assist.dla.mil -- Downloaded: 202610-16T06:47Z
Check the source to verify that this is the current version before use.

ment are met. If a product is manufactured to meet this spec

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Physical and chemical characteristics. The hexachloroethane shall conform to the physical and chemical characteristics of table I when tested as specified therein.

TABLE I. Physical and chemical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Melting point, °C	183.0	187.0	4.2.4.1
Assay (as hexachloroethane), percent by weight[1]	98.00	---	4.2.4.2
Acidity (as HCl), percent by weight	---	0.04	4.2.4.3
Iron, percent by weight	---	0.07	4.2.4.4
Water content, percent by weight	---	0.05	4.2.4.5
Ammonium chloride, percent by weight[2]	---	0.02	4.2.4.6

[1]Calculated from the total chlorine after making deductions for the chlorine in the hydrochloric acid.

[2]This requirement does not apply when the material is to be used in smoke mixtures containing ammonium chloride as one of the ingredients, or if the process used in the manufacture of hexachloroethane does not involve the use of ammonia.

3.1.1 Material. The hexachloroethane shall not be adulterated by the addition of any foreign material. The contractor shall certify that this requirement has been met.

Source: https://assist.dla.mil -- Downloaded: 20261-16T06:47Z
Check the source to verify that this is the current version before use.

3.2 Particle size. The hexachloroethane shall conform to the particle size characterisitics of table II when tested as specified in 4.2.4.7.

TABLE II. Particle size

Particle size	Percent by weight passing	
	Minimum	Maximum
850 micrometers	95	--
300 micrometers	50	--
150 micrometers	--	30
75 micrometers	--	10

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the hexachloroethane produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). When specified (see 6.2), lot numbering shall be in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. See 6.5 for sampling and testing precautions. Sampling shall be conducted in accordance with table III. A representative specimen of approximately 500 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:47Z
Check the source to verify that this is the current version before use.

TABLE III. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit container, ready for shipment. Sample unit containers shall be examined in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For critical defects. Each container in the lot shall be inspected for the critical defects listed in the classification of defects.

4.2.3.4 Classification of defects.

(a) Packaging (section 5).

Categories	Defects
Critical:	
1	Bag liner not heat sealed, tied, or knotted as specified
2	Container closure not secure
Major: AQL 1.0 percent defective	
101	Container not as specified
102	Bag liner not as specified
103	Quantity per container not as specified
104	Container damaged or leaking
105	Unitization not as specified
106	Marking incorrect, iliegible, or missing

4.2.4 Tests. See 6.5 for sampling and testing precautions. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determination shall be run and corrections applied where significant. Tests shall be conducted as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:47Z
Check the source to verify that this is the current version before use.

4.2.4.1 Melting point. Determine the melting point of the specimen in accordance with ASTM E324, except that the capillary tube shall be completely sealed approximately 15 millimeters from the specimen.

4.2.4.2 Assay. Determine the percent chloride of the specimen in accordance with ASTM E256. Calculate the percent by weight hexachloroethane as follows:

$$\text{Percent hexachloroethane} = 1.113A - 1.082B$$

where: A = Percent chlorine and
B = Percent hydrochloric acid (see 4.2.4.3).

4.2.4.3 Acidity. Determine the acidity of the specimen as percent by weight hydrochloric acid in accordance with ASTM D2989.

4.2.4.4 Iron.

(a) Butanolic potassium thiocyanate solution. Dissolve 10 g of potassium thiocyanate in 10 milliliters (mL) of water. Warm the solution to 25°C and add n-butanol to make 100 mL. Shake vigorously until the solution is clear.

(b) Ferric ammonium sulfate solution. Dissolve 0.1727 g of ferric ammonium sulfate in 10 mL of 10-percent sulfuric acid and dilute with water to make 2,000 mL. One mL of this solution equals 0.01 milligram (mg) of iron.

(c) Procedure. Weigh 7.00 g of the specimen into a large porcelain crucible. Add approximately 8 g of potassium bisulfate or potassium pyrosulfate, cover with a porcelain cover, heat gradually, and fuse until a clear melt is obtained. Cool to room temperature, add 25 to 50 mL of 1 to 4 hydrochloric acid, boil to dissolve the fusion, and filter. Wash two or three times with 1 to 3 hydrochloric acid. Add a few drops of hydrogen peroxide to the filtrate and then a slight excess of ammonium hydroxide. Boil until the ferric hydroxide forms flocks, cool to room temperature, filter through a fritted glass filter prepared with diatomaceous silica, and discard the filtrate. Attach the fritted glass filter containing the ferric hydroxide precipitate to a bell jar filtering apparatus. Add 10 mL of 1 to 1 nitric acid to the ferric hydroxide, filter, and catch the filtrate in a 250-mL beaker. Wash alternately with 1 to 1 nitric acid and hot water, using the minimum amount necessary to remove the ferric nitrate from the filter. Transfer the filtrate quantitatively to a 250-mL volumetric flask and dilute to 250 mL with water. Pipet a 2.5-mL aliquot into a test tube and add water to make a volume of approximately 10 mL. Add approximately 30 mg of ammonium persulfate and 15 mL of butanolic potassium thiocyanate solution prepared as specified in (a). Shake well for about 30 seconds and allow to separate. Any red color in the clear butanol (upper) layer shall be no darker than that in a blank in which 5 mL of ferric ammonium sulfate solution prepared as specified in (b) is substituted for the 2.5-mL aliquot and tested in the same manner.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:47Z
Check the source to verify that this is the current version before use.

4.2.4.5 Water content. Determine percent by weight water in the specimen in accordance with ASTM E203.

4.2.4.6 Ammonium chloride. Weigh to the nearest 0.01 g approximately 25 g of the specimen into a 500-mL separatory funnel. Add 100 mL of water and 100 mL of ether. Shake vigorously until two distinct layers have formed. Drain off the bottom water layer into a 500-mL Kjeldahl flask. Extract the ether layer twice more with 100-mL portions of water and transfer the water layers to the Kjeldahl flask. Discard the ether layers. Add a few drops of sulfuric acid to the Kjeldahl flask to make the solution distinctly acid and heat on a steam bath until only a trace of ether remains. Cool the flask to room temperature and then carefully add 25 mL of 50-percent sodium hydroxide solution down the side of the flask in such a manner that a separate layer is formed on the bottom of the flask. Immediately add to the flask several pieces of broken unglazed porcelain to promote uniform boiling and without agitation, attach the flask to the condenser of a Kjeldahl distillation apparatus. Extend the outlet tube of the condenser beneath the surface of a solution of 25.00 mL of 0.1N sulfuric acid and 25 mL of water in a 500-mL Erlenmeyer flask. Quickly agitate the contents of the Kjeldahl flask with a swirling motion and distill until approximately 150 mL have been distilled into the Erlenmeyer flask. Discontinue the distillation, disconnect the flask, and wash out the condenser with water, adding the washing to the distillate. Titrate the excess sulfuric acid with 0.1N sodium hydroxide solution using 6 drops of sodium alizarine sulfonate-indigo carmine indicator solution (0.9 g of sodium alizarine sulfonate and 0.125 g of indigo carmine in 100 mL of water). The end point is indicated by a change from green to bluish-gray. Calculate the percent by weight ammonium chloride as follows:

$$\text{Percent ammonium chloride} = \frac{5.35\ (AB - DC)}{W}$$

where: A = Milliliters of sulfuric acid used.
B = Normality of sulfuric acid,
C = Milliliters of sodium hydroxide solution used,
D = Normality of sodium hydroxide solution, and
W = Weight of specimen in grams.

4.2.4.7 Particle size. Place 100 ± 1 g of the specimen on a sieve stack arranged from top to bottom as follows: 850 micrometers, 300 micrometers, 150 micrometers, 75 micrometers, and bottom pan. Sieves shall conform to ASTM E11. Place the covered sieve stack in a mechanical shaker geared to produce 300 ± 15 gyrations per minute and 150 ± 10 taps of the striker per minute. Vibrate for 5 minutes. Weigh the specimen remaining on the sieves and in the pan and calculate the percent by weight passing through each sieve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:47Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Unit packing. Unit packing shall be level A, B or C as specified (see 6.2), and shall be in accordance with Department of Transportation (DOT) regulations and any other regulations applicable to the mode of transportation used. All shipments intended to be shipped by water transport shall be unit packed level A in accordance with DOT requirements.

5.1.1 Unit packing, level A. A quantity of 240 (+3 or -0) pounds (lb) of hexachloroethane shall be unit packed in a nominal 30-gallon (gal) capacity steel drum conforming to type IV of PPP-D-705. The drum shall be furnished with a polyethylene film bag liner conforming to type I, class B, style 2 of MIL-B-117. The bag shall be closed by tying, knotting or heat sealing. The drum shall be closed by bolting the locking ring, clamping the head to the drum while tapping the locking ring sharply. Start tapping at a point diametrically opposite the bolt and working left and right around the ring toward the bolt until the tapping indicates a firm seating of ring to drum.

5.1.2 Unit packing, level B. A quantity of 240 (+3 or -0) lb of hexachloroethane shall be unit packed level B in the same manner as specified for level A except that the drum shall conform to section 2, rule 51 of the Uniform Freight Classification (UFC) for contents that weigh over 225 lbs but not over 300 lbs, utilizing a sidewall construction including an aluminum barrier of 0.0005-inch minimum thickness as a component of one of the inner plies.

5.1.3 Unit packing, level C. A quantity of 100 (+1 or -0) lb of hexachloroethane shall be unit packed level C in a paper shipping sack conforming to a type optional, construction number 17 or 17X sack of UU-S-48, having moisture barrier protection MB4 next to the innermost wall.

5.2 Packing. Hexachloroethane, unit packed as specified in 5.1, shall require no further protection for shipment other than unitization.

5.3 Unitization. Uniform quantities of packs of hexachloroethane shall be palletized in accordance with the applicable requirements of MIL-STD-147, using the softwood pallet conforming to type IV of NN-P-71.

5.4 Marking. Packs of hexachloroethane shall be marked in accordance with MIL-STD-129 and DOT regulations. Each pack shall be marked to show the lot or batch number and the date of manufacture of the hexachloroethane. In addition, each container shall be durably and legibly marked with the following precautionary marking:

HAZARDS:

Irritant, Eye and Skin.
Toxic by Ingestion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:47Z
Check the source to verify that this is the current version before use.

WARNING! CAUSES IRRITATION
HARMFUL IF SWALLOWED
Avoid contact with eyes, skin, and clothing.
Wash thoroughly after handling.

FIRST AID: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Call a physician. Flush skin with water. (Wash clothing before reuse.)

If swallowed, induce vomiting by sticking finger down throat or by giving soapy or strong salty water to drink. Repeat until vomit is clear. Call a physician. Never give anything by mouth to an unconscious person.

SUPPLY/USE: Appropriate respiratory, eye and skin protection as prescribed by industrial hygiene, medical and safety authorities.

6. NOTES

6.1 Intended use. Hexachloroethane is intended for use in white smoke mixtures.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) If lot numbering in accordance with MIL-STD-1168 is required (see 4.2.1),
(c) Level of unit packing required (see 5.1), and
(d) If the ammonium chloride requirement is to be waived (see table I).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purposed of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Sampling and testing precautions. This specification covers inspection and use of chemical material which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material. Care should be taken in sampling the hexachloroethane and in handling the specimens to prevent loss of any volatile constituents. Containers should be kept securely closed and specimens should be handled as expeditiously as possible to prevent aeration of the hexachloroethane before tests are completed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:47Z
Check the source to verify that this is the current version before use.

Custodians:

Army - EA
Navy - AS

Review activities:

Army - MD
DLA - GS

User activities:

Army - AR
Navy - MC

Preparing activity:

Army - EA

Project No. 6810-B410

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:47Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for submitting comments and suggestions for improvements. All users of military standardization documents are invited to p suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)* mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretatio too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorizat to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO P
NEC
IF M
IN
UNITE

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON, D.C

POSTAGE WILL BE PAID BY DEPARTMENT OF THE ARMY

Commander
Armament Research and Development Center
US Army Armament, Munitions and Chemical Command
ATTN: DRSMC-TSC-S(A)
Aberdeen Proving Ground, MD 21010

Source: https://assist.dla.mil -- Downloaded: 202610-16T06:47Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-235C	HEXACHLOROETHANE, TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

EDITION OF 1 OCT 76 IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 202640-16T06:47Z
Check the source to verify that this is the current version before use.